A single leaky neuron: demonstration notes

Page 1: A neuron, at rest.

Background teaching material. Not a published research paper.

Before a neuron sends a signal, it holds a quiet electrical difference across its membrane.

In this model, that resting potential is -65 millivolts; without an input, the voltage stays there.

Parameters throughout these notes: rest/reset = -65 mV; resistance = 10 Mohm; default tau = 20 ms. A resting neuron starts at V = -65 mV. I is applied current in nA. These are illustrative model choices, not empirical measurements.

At rest with I = 0, both the input and the voltage difference from rest are zero. There is no net change, so the model remains at -65 mV.

Membrane: The boundary that holds the voltage difference.

Input current: The external drive into our model neuron, measured in nanoamps.

Axon: The output path that carries the neuron’s signal away from its cell body.

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Page 2: An input becomes a voltage.

Background teaching material. Not a published research paper.

A steady input raises the membrane potential, but the charge also leaks away.

The balance approaches a new resting level; the time constant controls how quickly it gets there.

Differential equation: tau * dV/dt = -(V - Vrest) + R*I. Numerical method: forward Euler, dt = 0.1 ms. At each step, V += (-(V+65)+10*I)/tau * 0.1. Equilibrium: Vrest + R*I. There is no stochastic input.

The time constant divides the voltage’s rate of change. A larger tau slows the approach, while the equilibrium Vrest + RI remains the same. With 1 nA and R = 10 Mohm, that equilibrium is -55 mV.

Leak: The tendency for the membrane voltage to move back toward rest.

Integration: Input and leak together determine how voltage changes over time.

Steady input: A constant current pushes the voltage toward Vrest + RI.

Axon: The output path that carries the neuron’s signal away from its cell body.

PAPER-TO-WORLD / AUTHORED DEMONSTRATION NOTES

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Page 3: A threshold changes everything.

Background teaching material. Not a published research paper.

When the membrane reaches -50 mV, our model emits a spike and resets to -65 mV.

A two-millisecond refractory interval follows, then the membrane begins integrating again.

After the Euler voltage update, if V >= -50 mV: increment spike count; set V = -65 mV; set refractory timer = 2 ms. While that timer is positive, decrement it by dt (clamped at zero) and hold V = -65 mV. A +20 mV graphical marker records an event, not an action-potential waveform.

An input of 1 nA approaches -55 mV, below the -50 mV threshold. Inputs greater than 1.5 nA can reach the threshold in this model. The spike marker at +20 mV is a drawing convention, not a biological action-potential waveform.

Threshold: The voltage at which the model emits a spike: -50 mV.

Spike: A discrete output event, drawn as a sharp line in the trace.

Reset: After a spike, voltage returns to -65 mV before integration resumes.

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Page 4: More input. A faster rhythm.

Background teaching material. Not a published research paper.

Stronger input shortens the journey to threshold, so the neuron fires more often.

Our simulation produced 33 spikes at 2 nA and 63 spikes at 3 nA in the first 1 s (0?1000 ms); the tags and the plot count the same window.

Reference targets measured over 1000 ms at tau=20 ms: I=0 nA -> 0 spikes; I=1 nA -> 0 spikes; I=2 nA -> 33 spikes; I=3 nA -> 63 spikes. Approximate continuous-model rate for I>1.5 nA: f=1000/(tau*ln(R*I/(R*I-15))+2) Hz. Finite observation time and discrete Euler updates cause small differences from this formula. Passing numerical agreement is not validation against a published paper or biological data.

The charging time depends on a logarithm, and every spike adds a refractory interval. Our simulation gives 33 spikes at 2 nA and 63 at 3 nA in the first 1 s (0?1000 ms), matching the checked Python reference. This is numerical agreement with our reference model, not validation against a research paper or biological data. The analytic expression is approximate for the discrete model.

Firing rate: Spikes counted in the first 1 s (0?1000 ms) of a run, per second.

Interspike interval: Time between two spike events; shorter intervals mean a faster rhythm.

Threshold: The same -50 mV boundary, reached faster under stronger input.